Statement of Work

Title: AI Concept Visualizer and Explainer

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Background and Motivation

TikTok and YouTube Shorts have shown how engaging bite-sized content can be when it combines visuals and audio. What if we learned the same way? We know first-hand the challenges of breaking down abstract or technical material, and we believe there are better ways to communicate, learn, and understand complex ideas.

We have also seen the Manim animation library successfully used to manually create visual educational explainers. At the same time, generative AI models have made it possible to automatically generate TikTok-style audio and visuals. By combining these, we aim to explore how AI can be used to generate short-form visual explainers for complex topics such as math, physics, and machine learning (ML), with the potential to extend to other areas.

Problem Statement:

Develop an application that automatically generates short-form, visually engaging explainers for technical concepts using generative AI and Manim animations.

Data Sources:

We will first attempt to rely on the LLM's internal knowledge to generate Manim code, avoiding the overhead of retrieval. If we encounter accuracy issues or hallucination, we will incorporate RAG using official Manim and plugin documentation to ground outputs.

Source: Manim and Manim plugin documentation (official docs, API references, examples).
Description: Full-text access to docs, and code samples used as grounding for animation generation.
Key Attributes: Versioned and searchable, includes API signatures and usage patterns.
Relevance: Ensures the pipeline produces correct Manim code and reduces hallucination.
Data Quality Concerns: Doc/version drift across Manim releases, we will pin versions and validate against local Manim builds.

Scope and Objectives:

AI-generated visual explainers for complex concepts + chat-style interface for video generation.

Objectives:

1. **Video Generation Workflow:** Develop a LangGraph agent pipeline that converts AI-generated scripts into short videos with narration and Manim-based animations. We may optionally add the Model Context Protocol (MCP) for modularity.
2. **Model Hosting Strategy:** Likely we will run models locally through Ollama instead of using cloud APIs. This approach would minimize cost and latency by running everything on the user's machine, while the other approach would enable a purely web-based app.

3. **Proof-of-Concept Demos:** Create sample explainers in math, physics, and ML.
4. **User Interface:** Build a chat-style interface for prompts, script/storyboard previews, and approval before full video generation. Optional extensions: captions and audio toggles.
5. **Cloud Deployment:** Deploy a minimal backend cloud service (e.g., on Render, Fly.io, or Vercel) to handle user login/authentication, and project metadata.
6. **Evaluation:** Assess (a) factual accuracy of generated content, (b) clarity and engagement, based on informal peer feedback, compared to more traditional formats, and (c) adaptability across different subject areas without extensive manual editing.

Minimum Components for a Good Project:

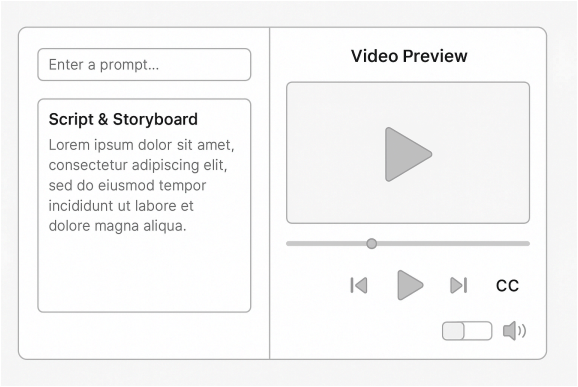
1. **Large Data:** User prompts, AI-generated scripts, and Manim documentation for grounding.
2. **Scalability:** Backend must support multiple users simultaneously.
3. **Complex Models:** LLMs for script generation and text-to-speech for narration.
4. **Computationally Expensive Inference:** Optimize rendering and inference to minimize latency.

Learning Emphasis:

The project emphasizes AI workflow integration: natural language generation for scripts, text-to-speech for narration and Manim for animations.

Application Mock Design:

- **Interface 1:** Chat-style interface for entering prompts, previewing scripts/storyboards, and approving content.
- **Interface 2:** Video preview interface showing generated explainers with optional captions and audio toggles.



Research and Development:

1. ArXive TikToks: https://glitter-loaf-f80.notion.site/ArXive-TikTok

Fun Factor:

This project combines AI with education, turning complex topics into bite-sized, interactive, and shareable learning experiences that feel more engaging and accessible.

Milestones:

1. *(Sep. 28) Data Acquisition:* Collect Manim and plugin documentation and provide RAG to LLM.
2. *(Oct. 12) LangGraph:* Build agent and UI to take user prompts, generate Manim code.
3. *(Oct. 26) Animation:* Render generated Manim code, add option to save video locally.
4. *(Nov. 9) Narration:* Generate narration script, implement TTS aligning with video for playback.
5. *(Nov. 30) Accounts:* User authentication flow, connecting accounts to backend and saving videos.
6. *(Dec 9) Deployment:* Deploy application.
7. *(Throughout) Demos:* At each stage in the process, create demos for the final presentation.